

BIO202 BIOCHEMISTRY-I MIDTERM MEGA FILE

ALL DATA FROM PAST PAPERS 2017-2022

GROUP: BIOTECH BRAINY BUNCH

ADMINS: MEHWISH MUGHAL & BELLA

CONTACT: mehwishshabirhussain07@gmail.com

1) Write characteristics of enantiomers?

Stereoisomers that are mirror images of each other are called enantiomers.

Enantiomers have nearly identical chemical properties but differ in a characteristic physical property: their interaction with plane-polarized light. Since they are non-superposable mirror images of each other, the two forms represent a class of stereoisomers called enantiomers.

2) Name two diseases caused by mis-folding of proteins?

Mis-folding of proteins may occur spontaneously or caused by a mutation in a particular gene, producing an altered protein. In addition, some apparently normal proteins can, after abnormal proteolytic cleavage, take on a unique conformational state that leads to the formation of long fibrillary protein assemblies consisting of β -pleated sheets.

3) What are digestive proteins. Explain with examples?

Digestive Proteins: These proteins are digestive enzymes which digest our food materials such as carbohydrates, proteins, lipids and include Amylase, Pepsin, Lipases etc. (Pepsin is one of three principal protein-degrading, or proteolytic, enzymes in the digestive system, the other two being chymotrypsin and trypsin).

4) What are ketogenic amino acids?

Ketogenic amino acids: Amino acids whose catabolism yields either acetoacetate or its precursor, (acetyl CoA or aceto-acetyl CoA) are termed ketogenic. Acetoacetate is one of the ketone bodies which also include 3-hydroxybutyrate and acetone. Leucine and lysine are the only exclusively ketogenic amino acids found in proteins. Their carbon skeletons are not substrates for gluconeogenesis and, therefore, cannot give rise to the net formation of glucose or glycogen in the liver, or glycogen in the muscles.

5) Give characteristics of furfural compounds?

Furfural is an important renewable, non-petroleum based, chemical feedstock.) The furfural products thus formed can condense with certain organic phenols (a-naphthol) to form compounds having characteristic colors This forms basis of certain tests used for detection of sugars

6) Name factors that influence free energy of protein molecules?

The free energy of a protein molecule is influenced by (1) the hydrophobicity, hydrogen bonds, electrostatic interactions, and (4) the conformational entropy due to the restricted motion

07) what do you know about allosteric amino acid?

In biochemistry, allosteric regulation (or allosteric control) is the regulation of an enzyme by binding an effector molecule at a site other than the enzyme's active site. The site to which the effector binds is termed the allosteric site or regulatory site. Allosteric sites allow effectors to bind to the protein, often resulting in a conformational change involving protein dynamics. Effectors that enhance the protein's activity are referred to as allosteric activators, whereas those that decrease the protein's activity are called allosteric inhibitors.

08) Define allotropy?

the existence of two or more different physical forms of a chemical element

09) what is HSP60?

Heat shock protein 60 (HSP60) is a mitochondrial chaperonin that is typically held responsible for the transportation and refolding of proteins from the cytoplasm into the mitochondrial matrix.

10) What are characteristics of furfural compounds

Furfural is an organic compound produced from a variety of agricultural byproducts, including corncobs, oat, wheat bran, and sawdust. The name furfural comes from the Latin word furfur, meaning bran, referring to its usual source. Furfural is an almond-scented, oily, colorless liquid that turns yellow to dark brown when exposed to air. It is used as a solvent for refining lubricating oils, as a fungicide and weed killer and in the production of tetrahydrofuran, an important industrial solvent.

11) How Duchenne Muscular Dystrophy is caused?

DMD is a genetic disease. Those who inherit it have a defective gene related to a muscular protein called dystrophin. This protein keeps muscle cells intact. Its absence causes rapid muscular deterioration as a child with DMD grows.

12) Difference between Hemiacetals and Hemiketals?

The formation of these ring structures is the result of a general reaction between alcohols and aldehydes or ketones to form derivatives called hemiacetals or hemiketals. Which contain an additional asymmetric carbon atom and thus can exist in two stereoisomeric forms

13) How mis-folding of proteins occur?

Protein misfolding: aggregation and amyloid formation. Protein misfolding is a common and intrinsic propensity of proteins that occurs continuously. Misfolding is influenced by the amino acid composition, and certain mutations are known to accelerate the process.

14) What are monosaccharides? write two examples.

Monosaccharides (simple sugars) are those which cannot be hydrolyzed further into simpler forms. The backbones of common monosaccharides are unbranched carbon chains in which all the carbon atoms are linked by single bonds.

15) What are Chaperons and write its main function?

Molecular chaperones are proteins that assist the covalent folding or unfolding and the assembly or disassembly of other macromolecular structures. The reason for this behavior is that protein folding is severely affected by heat and, therefore, some chaperones act to prevent or correct damage caused by misfolding. Other chaperones are involved in folding newly made proteins as they are extruded from the ribosome.

16) What are anomers and anomeric carbon?

The anomeric carbon is a stereo center. An important feature is the direction of the OH group attached to the anomeric carbon. When the hydrogen is still attached, and the oxygen is not bonded C-O-C, this is called a hemiacetal functional group.

17) What is starch and its structure?

Starch molecules arrange themselves in the plant in semi-crystalline granules. Each plant species has a unique starch granular size: rice starch is relatively small (about 2 µm) while potato starches have larger granules (up to 100 µm). Starch becomes soluble in water when heated.

18) What are beta bonds?

1,4-glycosidic bond. ... There are two types of glycosidic bonds - 1,4 alpha and 1,4 beta glycosidic bonds. 1,4 alpha glycosidic bonds are formed when the OH on the carbon-1 is below the glucose ring; while 1,4 beta glycosidic bonds are formed when the OH is above the plane.

19) What side chain importance of alanine, valine, leucine?

Role in structure: Alanine is arguably the most boring amino acid. It is not particularly hydrophobic and is non-polar. ... Role in function: The Alanine side chain is very non-reactive, and is thus rarely directly involved in protein function. Valine is a branched-chain amino acid (BCAA) that works with the other two BCAAs, isoleucine and leucine, to promote normal growth, repair tissues, regulate blood sugar, and provide the body with energy. Valine helps stimulate the central nervous system, and is needed for proper mental functioning. Previous studies with Substance P (SP) antagonists (GR 71251, pPro9, Pro10, Trp1 11SP and [DPro9, MeLeu1 0, Trp1 1]SP) have suggested the existence in the guinea-pig ileum (GPI) of two distinct tachykinin receptors associated with the contractile responses of pro91SP and septide.

20) What are domain in protein?

A protein domain is a conserved part of a given protein sequence and (tertiary) structure that can evolve, function, and exist independently of the rest of the protein chain. Each domain forms a compact three-dimensional structure and often can be independently stable and folded.

21) What are protein strictures?

The primary structure of a protein refers to the sequence of amino acids in the polypeptide chain. The primary structure is held together by peptide bonds that are made during the process of protein biosynthesis.

22) What are basics of classification of proteins?

Protein Classification. Proteins are the macromolecules responsible for the biological processes in the cell. They consist at their most basic level of a chain of amino acids, determined by the sequence of nucleotides in a gene. ... A polypeptide is a chain of amino acids.

23) Write a note on transcription?

Transcription is the process of producing an RNA transcript that is complementary to a DNA template. It can be thought of as DNA-dependent RNA synthesis.

24) what is disaccharide give two examples

A disaccharide (also called a double sugar or blase) is the sugar formed when two monosaccharides (simple sugars) are joined by glycosidic linkage. Like monosaccharides, disaccharides are soluble in water. Three common examples are sucrose, lactose, and maltose.

25) What is quaternary structure of protein?

Many proteins are actually assemblies of multiple polypeptide chains. The quaternary structure refers to the number and arrangement of the protein subunits with respect to one another. Examples of proteins with quaternary structure include hemoglobin, DNA polymerase, and ion channels.

26) how alpha helix stabilize?

The alpha helix is a rod like structure. A tightly coiled backbone forms the inner part of the rod and the side chain extends outwards in a helical manner. The alpha helix is stabilized by hydrogen bonds between the NH and CO groups of the main chain i.e the CO group of each amino acids forms a H-bond with the NH group of the amino acid i.e situated four residues ahead in the sequence

27) What is biological importance of carbohydrates?

Carbohydrates provide energy to the body, particularly through glucose, a simple sugar that is found in many basic foods. ... As an immediate source of energy, glucose is broken down during the process of cellular respiration, which produces ATP, the energy currency of the cell.

28) what is role proton bind with heme group in tissues and lung?

Protons bind at various places on the protein, while carbon dioxide binds at the α -amino group. ... A reduction in the total binding capacity of hemoglobin to oxygen (i.e. shifting the curve down, not just to the right) due to reduced pH is called the root effect. Function Tissue and Lungs: Hemoglobin transport O₂ from lungs to tissues and CO₂ from tissues to lung.

29) What are protein structures?

Protein structure is the three-dimensional arrangement of atoms in an amino acid-chain molecule. Proteins are polymers — specifically polypeptides — formed from sequences of amino acids, the monomers of the polymer. A single amino acid monomer may also be called a residue indicating a repeating unit of a polymer. Proteins form by amino acids undergoing condensation reactions, in which the amino acids lose one water molecule per reaction in order to attach to one another with a peptide bond. By convention, a chain under 30 amino acids is often identified as a peptide, rather than a protein.

30) Example of protein denaturation in our daily life?

Examples of Denatured Proteins. Though protein denaturation is detrimental for cell survival, it is often encountered in daily life. ... Similarly, altering the pH of milk by adding acids such as citric acid from lemon juice denatures milk proteins and curdles the milk.

31) what are ketogenic amino acid?

Ketogenic amino acid. A ketogenic amino acid is an amino acid that can be degraded directly into acetyl-CoA, which is the precursor of ketone bodies. This is in contrast to the glucogenic amino acids, which are converted into glucose.

32) describe alpha helix or disulphide bridge?

It is a sequence of amino acids. 2. It is a linear polymer: linking the alpha-carboxyl group of one amino acid to the alpha amino group of another amino acid => PEPTIDE BOND (covalent bond). 3. In some proteins, the linear polypeptide chain is cross-linked: Disulfide bonds. The primary structure is a polypeptide, in which:

+ each amino acid in the peptide is a residue. + there is a regularly repeating segment called the main chain or backbone, and a variable part, comprised of the side chain.

33) What are allosteric form of hemoglobin?

Hemoglobin is an allosteric protein. ... In fact, the binding of oxygen to one hemoglobin subunit induces conformational changes (discussed before) that are relayed to the other subunits, making them abler to bind oxygen by raising their affinity for this molecule.

34) What is titration curve of glycine?

The end point of a titration curve represents the observed end of the titration. ... In this experiment we are finding out the titration curve of the amino acid Glycine. Glycine is a diprotic amino acid which means that it has two dissociable Protons, one on an amino group and the other on the carboxyl group.

35) What are Motif of protein?

In a chain-like biological molecule, such as a protein or nucleic acid, a structural motif is a super secondary structure, which also appears in a variety of other molecules. Motifs do not allow us to predict the biological functions: they are found in proteins and enzymes with dissimilar functions.

36) What is Equilibrium constant of reversible reaction?

There is a simple relationship between the equilibrium constant for a reversible reaction and the rate constants for the forward and reverse reactions if the mechanism for the reaction involves only a single step. ... This equation can be rearranged to give the equilibrium constant expression for the reaction.

37) Difference btw secondary and tertiary structure of proteins

Explain the differences between primary, secondary and tertiary protein structure. Primary structure is the order in which what amino acid is bound the other with a peptide bond. This is coded for by the order of codons in a gene. ... This structure is determined by hydrogen bonds between the different amino acids

38) Define electrostatic forces in protein interactions?

Protein electrostatic properties stem from the proportion and distribution of polar and charged residues. Polar and charged residues regulate the electrostatic properties by forming shod-range interactions, like salt-bridges and hydrogen-bonds, and by defining the over-all electrostatic environment in the protein. Electrostatics play a major role in defining the mechanisms of protein-protein complex formation, molecular recognitions, thermal stabilities, conformational adaptabilities and protein movement

39) Domain in Protein and their function?

A protein domain is a conserved part of given protein sequence and (tertiary) structure that can evolve, function, and exist independently of the rest of the protein chain. ... Many proteins consist of several structural domains. One domain may appear in a variety of different proteins.

40) Explain the single letter naming symbol in Amino acids?

SHORTHAND SYMBOLS FOR AMINO ACIDS

[1-letter symbols are commonly used in sequence data]

One letter	Three letter	Amino Acid
A	Ala	Alanine

R	Arg	Arginine
N	Asn	Asparagine
D	Asp	Aspartic acid

42) Role of Hsp60?

The Hsp60 class of protein has a barrel shape into which the protein fits. It acts as a template, binding and rebinding portions of the unfolded protein until folding is completed. It hydrolyzes many ATP bonds to provide energy for the process.

43) Write the name of Essential amino acids?

here are 10 essential amino acids: arginine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine.

44) What are properties of proteins?

Proteins are compounds found in all living cells, in animals and plants. They play a variety of important roles and are essential to maintain the structure and function of all lifeforms. The word 'protein' is derived from the Greek word protons, meaning "primary" or "first". Proteins are vital for the growth and repair, and their functions are endless. Each and every property that characterizes a living organism is affected by proteins, whether it is a bacteria or a human body hydrolysis.

Protein s - Peptides - Amino acids.

45) discuss the steps of protein folding?

There are four stages of protein folding, primary, secondary, tertiary and quaternary. The secondary structure is the protein beginning to fold up. It can have two types of structure: the alpha helix, a coil shape held by hydrogen bonds in the same direction as the coil.

46) what are osazone?

Osazones are a class of carbohydrate derivatives found in organic chemistry formed when sugars are reacted with excess of phenyl hydrazine. The famous German chemist Emil Fischer developed and used the reaction to identify sugars whose stereochemistry differed by only one chiral carbon

47) What are monosaccharides equation reaction?

Monosaccharide is simple sugar and are not hydrolyzed further into simpler forms. They are unbranched and are in open chain form. examples are glucose, fructose, ribose

48) Tryptophan, Phenyl alanine, and Tyrosine Structure?

Tryptophan: The more complex ring structure in tryptophan is an indole ring with a nitrogen that can engage in hydrogen bonds. Tryptophan is therefore also more polar than phenylalanine. Tyrosine and tryptophan are significantly more polar than phenylalanine, because of the tyrosine hydroxyl group and the nitrogen of the tryptophan indole ring.

Phenylalanine: In the amino acid phenylalanine, the ring contains no substituents, and the electrons are shared equally between the carbons in the ring, resulting in a very nonpolar.

Tyrosine: a hydroxyl group on the phenyl ring engages in hydrogen bonds, and the side chain is therefore more polar and more hydrophilic.

49) Which enzymes convert maltose into sugar?

The enzyme maltase present in the lining of the small intestine breaks maltose into two units of glucose.

50) Lactose intolerance?

Lactose intolerance is a condition in which people have symptoms due to the decreased ability to digest lactose, a sugar found in dairy products. ... Symptoms may include abdominal pain, bloating, diarrhea, gas, and nausea.

51) Write 2 Name Charged R groups?

Lysine, arginine, histidine

52) Carbohydrate and its biological importance?

Carbohydrates are all about energy and are found in foods like fruits, vegetables, breads, pasta, and dairy products. Your body uses these foods to make glucose, which is your body's main energy source. Glucose is a type of sugar that can be used right away for energy or stored away to be used later.

53) Alanine Structure?

Alanine is an α -amino acid that is used in the biosynthesis of proteins. It contains an α amino group, an α -carboxylic acid group, and a side chain methyl group, making it a nonpolar, aliphatic amino acid.
Formula: $C_3H_7NO_2$

54) Amino acid sequence?

Protein molecules are made of strings of amino acids in a particular order. This string is called an amino acid sequence

55) Sickle cell anemia?

In Sickle cell anemia a Glutamate residue is replaced by Valine at position 6 of beta chain of Hemoglobin, resulting in decreased functioning and increased fragility of Hb.

56) Factor of influence free energy of protein?

The free energy of a protein molecule is influenced by:

- the hydrophobicity
- hydrogen bonds,
- electrostatic interactions

- the conformational entropy due to the restricted motion

57) Protein involve in respiration?

This produces the sweet smell of acetone on the breath, noticeable in a diabetic state. Proteins are used as an energy source only if protein intake is high, or if glucose and fat sources are depleted, in which case amino acids from protein breakdown are converted into molecules that can enter the TCA Cycle.

58) Alanine curve and value?

On the left in both the chemical reaction and the titration curve, you should imagine that alanine is in a very acidic solution at a pH of about 0. As you move from left to right across the page, you are adding hydroxide to the solution. This increases the hydroxide concentration and decreases the hydrogen ion concentration. Note that alanine loses protons as you move from left to right. At the first pKa, the alpha-carboxyl dissociates. At the second pKa, the alpha amino dissociates. IP Value is 5.7.

59) Explain Electrostatic interaction in protein?

Specific interactions are largely electrostatic. While the hydrophobic effect is the major driving force in protein folding, electrostatic interactions are important in protein folding, stability, flexibility, and function.

60) Explain term induce fit?

The structural adaptation that occurs between protein and Ligand is called induced fit.

61) What is anomers?

Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called anomers

62) Name the factors that counteract entropy?

The hydrophobic effect and hydrogen bonds are the major stabilizing contributions which counteract entropy.

63) What is the mean by equilibrium constant for reversible reaction?

The tendency of any acid (HA) to lose a proton and form its conjugate base (Ave-) is defined by the equilibrium constant (K_{eq}) for the reversible reaction

64) What are the perk quaternary structure?

The increase in size increases the number of possible interactions between amino acid residues and therefore makes it more difficult for a protein to unfold and refold.

65) Amino Acid?

A simple organic compound containing both a carboxyl (—COOH) and an amino (—NH₂) group.

66) Aliphatic non polar R group?

Aliphatic R groups are nonpolar and hydrophobic. Hydrophobicity increases with increasing number of C atoms in the hydrocarbon chain. Although these amino acids prefer to remain inside protein molecules, alanine and glycine are ambivalent, meaning that they can be inside or outside the protein molecule.

67) Glycine and proline why exist in beta sheet?

Glycine because it is small and flexible. Proline because peptide bonds involving the imino nitrogen readily assume the cis configuration, a form that is particularly amenable to a tight turn.

68) PKA

The pKa expresses, on a logarithmic scale, the relative strength of a weak acid or base

69) Why proline disrupt alpha helix?

In proline, the nitrogen atom is part of a rigid ring, and rotation about the N-C bond is not possible. Thus, a proline residue introduces a destabilizing kink in an α helix.

70) What is Keratin?

Keratin are a family of closely related, fibrous proteins whose structure is nearly entirely α -helical. They are a major component of tissues such as hair, nails, hoofs and skin. In contrast to keratin, myoglobin, whose structure is also highly α -helical, is a globular, flexible molecule. Nearly all trans membrane proteins contain alpha helices in their membrane spanning domains.

71) IP Value of lysine?

The IP value of lysine is 9.47

72) Function of heme in hemoglobin and myoglobin?

Functional relation of each protein in the subsequent discussion to see how both these proteins are suited to perform their required function in their relative sites in the body.

73) Glutamate and Glutamine write one letter representation?

Glutamate (Glu), Glutamine (Gln)

74) Define isomerism?

Isomerism is the phenomenon whereby certain compounds, with the same molecular formula, exist in different forms owing to their different organizations of atoms.

75) Glucogenic amino acid?

Thirteen amino acids: glycine, alanine, serine, aspartic acid, asparagine, glutamic acid, glutamine, proline, valine, methionine, cysteine, histidine and arginine, are purely 'glucogenic'

76) Name Five protein product?

Eggs. Greek Yogurt. Cottage Cheese. Swiss Cheese. 2-Percent Milk

77) D amino acid?

D-amino acids are also non-standard amino acids that occur naturally and include –free D- serine, and D-aspartate in brain tissue –D-alanine & D-glutamate in the cell walls of gram positive bacteria. D-amino acids are also found in some antibiotics.

78) Optical activity isomers?

The two isomers non-superimposable mirror images of each other, we call them optical isomers or enantiomers. An example is the amino acid alanine

79) Polysaccharide?

A carbohydrate whose molecules consist of a number of sugar molecules bonded together. (e.g. starch, cellulose, or glycogen).

80) Carbohydrate and Major Groups?

A carbohydrate is a biomolecule consisting of carbon, hydrogen and oxygen atoms, usually with a hydrogen–oxygen atom ratio of 2:1.

- Monosaccharides: They are the simplest form of sugar and are usually colorless, water soluble, and crystalline solids. Some monosaccharides have a sweet taste. Examples of monosaccharides include glucose (dextrose), fructose (levulose), and galactose.
- Polysaccharide: A carbohydrate whose molecules consist of a number of sugar molecules bonded together. (e.g. starch, cellulose, or glycogen).
- Oligosaccharide: a carbohydrate whose molecules are composed of a relatively small number of monosaccharide units.
- Disaccharide: any of a class of sugars whose molecules contain two monosaccharide residues.

81) Hydrophobic interaction?

Hydrophobic Interactions occur between non-polar uncharged side chains such as between aromatic groups of phenylalanine side chains.

82) Disaccharide and two example?

A disaccharide (also called a double sugar or biose) is the sugar formed when two monosaccharides (simple sugars) are joined by glycosidic linkage. Like monosaccharides, disaccharides are soluble in water. Three common examples are sucrose, lactose, and maltose.

83) D and L Isomer?

The designation of a sugar isomer as the D form or as the L form is determined by its spatial relationship to the parent compound of the carbohydrates, the glycer aldehydes.

84) Tautomer?

Each of two or more isomers of a compound which exist together in equilibrium, and are readily interchanged by migration of an atom or group within the molecule.

85) Protein disulphide isomers?

Protein disulfide isomerase, or PDI, is an enzyme in the endoplasmic reticulum (ER) in eukaryotes and the periplasm of bacteria that catalyzes the formation and breakage of disulfide bonds between cysteine residues within proteins as they fold.

86) Disulphide bridge in tertiary structure?

The amino acid cysteine in a protein can form a covalent disulfide bond with another cysteine molecule through spontaneous (nonenzymatic) oxidation of their sulfhydryl groups.

87) Hemoglobin is found in?

Hemoglobin is found in the red blood cells of all vertebrates apart from white-blooded fish. It also occurs in some invertebrates.

88) Net Charge on alanine?

Net Charged on alanine is Zero (isometric form).

89) Monosaccharide and two example?

They are the simplest form of sugar and are usually colorless, water-soluble, and crystalline solids. Some monosaccharides have a sweet taste. Examples of monosaccharides include glucose (dextrose), fructose (levulose), and galactose.

90) Function of lanolin?

Lanolin and its derivatives are used in the protection, treatment and beautification of human skin.

91) Enzyme and classification?

a substance produced by a living organism which acts as a catalyst to bring about a specific biochemical reaction. Enzymes are classified according to the reactions they catalyze. The six classes are as follows: oxidoreductases, transferases, hydrolases, lyases, isomerases, and ligases

92) Waxes? properties and types?

Waxes A second group of neutral lipids that are of physiological importance. Although they are a minor component of biological systems. Properties: Waxes are insoluble in water, but soluble in fat solvents and are negative for acrolein test. very resistant to rancidity. Waxes are not easily hydrolyzed as the fats

and are indigestible by lipases (enzymes responsible for fat digestion in body) Thus they are of no nutritional value.

Waxes are of two types:

- True waxes (Bees-wax, • Spermaceti of the sperm whale)
- Other Waxes or Non true waxes or Wax-like compounds.

93) Renature and Explain with ribonuclease?

Regain their native structure and their biological activity if returned to conditions in which the native con-formation is stable.

- At Very High or Low ph.
- At Very High Temperatures.
- By Heavy Metal Ions.
- By Small Polar Molecules

This process is called renaturation. A classic example is the denaturation and renaturation of ribonuclease. A Purified Ribonuclease A denatures completely in a concentrated urea solution in the presence of a reducing agent. When the urea and the reducing agent are removed, the randomly coiled, denatured ribonuclease spontaneously refolds into its correct tertiary structure, restoring its catalytic activity. This proves that denaturation of some proteins is reversible.

94) What are prosthetic groups?

Prosthetic Group. Prosthetic group is the same as the coenzyme but are tightly bound to the enzyme. When they are split off, the enzyme is mostly denatured. Examples include flavin nucleotides and heme.

95) Define the structure of amino acid?

All amino acids found in proteins have this basic structure, differing only in the structure of the R-group or the side chain. The simplest, and smallest, amino acid found in proteins is glycine for which the R-group is a hydrogen (H). In proteins, only the L-isomer is found normally.

96) Why glycine does not form alpha helix?

Glycine occurs infrequently in α helices for a different reason. It has more conformational flexibility than the other amino acid residues. Polymers of glycine tend to take up coiled structures quite different from an α helix.

97) Destabilized the alpha helix?

Charge amino acid side chains have a tendency to destabilize the alpha helical or beta pleated sheet structures.

98) Define buffer

A buffer solution is an aqueous solution consisting of a mixture of a weak acid and its conjugate base, or vice versa. Its pH changes very little when a small amount of strong acid or base is added to it.

99) Epimer and its two example

Each of two isomers with different configurations of atoms about one of several asymmetric carbon atoms present. E.g The sugars glucose and galactose are epimers

100) Medical important of protein?

Protein is an important component of every cell in the body. ... Your body uses protein to build and repair tissues. You also use protein to make enzymes, hormones, and other body chemicals. Protein is an important building block of bones, muscles, cartilage, skin, and blood.

101) Catalytic Protein?

Enzyme catalysis is the increase in the rate of a chemical reaction by the active site of a protein. The protein catalyst (enzyme) may be part of a multi-subunit complex

102) Aromatic amino acid and example?

The aromatic amino acids have been grouped together because they all contain ring structures with similar properties, but their polarity differs. Example Phenylalanine, Tyrosine, Tryptophan.

103) Role hydrogen bond in histidine?

Hydrogen Bonds Involving Histidine Imidazole Nitrogen Atoms: A New Structural Role for Histidine Residues in Proteins. ... The amino acid histidine can play a significant role in the structure and function of proteins.

104) Factor Stability of protein?

Factors that stabilize the protein are:

- PH
- Temperature.

105) Amylase?

An enzyme, found chiefly in saliva and pancreatic fluid, that converts starch and glycogen into simple sugars.

106) What is Honey?

Honey is largely invert sugar and the presence of fructose accounts for the greater sweetness of honey. They produce an enzyme called invertase in their salivary glands. Enzymes are organic compounds that speed up a biochemical reaction.

107) Reversible binding of protein?

The functions of many proteins involve the reversible binding of other molecules. • A molecule bound reversibly by a protein is called a ligand.

108) Isoelectric point?

The isoelectric point is the Ph at which a particular molecule carries no electrical charge

109) Tautomerization

The process of shifting a hydrogen atom from one carbon atom to another to produce enediols is known as tautomerization Sugars possessing anomeric carbon atom undergo tautomerization in alkaline solutions • When glucose is kept in alkaline solution for several hours, it undergoes isomerization to form D-fructose and D-mannose

110) What is Bohr effect?

Bohr effect is a phenomenon first described in 1904 by the Danish physiologist Christian Bohr. Hemoglobin's oxygen binding affinity is inversely related both to acidity and to the concentration of carbon dioxide. it also means the reciprocal coupling of proton and oxygen binding termed as Bohr effect.

111) Binding of oxygen with hemoglobin at sea level as well as at high altitude?

The amount of hemoglobin in blood increases at high altitude. This is one of the best-known features for hemoglobin. Increasing the amount of hemoglobin in the blood increases the amount of oxygen that can be carried. so, at high altitudes the atmospheric pressure as well as the partial pressure of oxygen is lower than that at the ground level. This leads to low concentration of the oxygen in the blood. For this reason, binding affinity decreases.

112) Differentiate between enantiomer and diastereomer?

- Enantiomers are the chiral molecules that are mirror images of one another and are not superimposable.
- Diastereomers are the stereomer compounds with molecules that are not mirrored images of one another and that are not super imposable

113) What are alpha beta and gamma carbon?

The alpha carbon in organic molecules refers to the first carbon atom that attaches to a functional group, such as a carbonyl. The second carbon atom is called the beta carbon and the carbon present next to the beta carbon is the gamma carbon.

114) How A and B antigen differ from O-antigen?

Blood group A – has A-antigens on the red blood cells with anti-B antibodies in the plasma. blood group B – has B-antigens with anti A-antibodies in the plasma. ... blood group AB – has both A and B antigen and

have no antibodies A and B. Blood group O has no antigens, but they contain both anti-A and anti-B antibodies in the plasma.

115) heme interaction.

Heme synthesis starts in mitochondria with the condensation of succinyl-CoA with the amino acid glycine, activated by pyridoxal phosphate. ALA synthase is the rate-limiting enzyme of heme synthesis. Heme-heme interaction consists of a change of tension at the hemes, caused by a transition between two alternative quaternary structures of the protein.

Dr Perutz describes how spin changes that accompany reaction with ligands alter the oxygen affinity of the hemes. The cooperative interaction between dioxygen binding sites that occurs when dioxygen binds to hemoglobin; as oxygenation proceeds, the combination of additional dioxygen molecules is made easier.

116) Selenocysteine

the 21st amino acid, exists naturally in all kingdoms of life as the defining entity of seleno proteins. Functions: Selenocysteine is an analogue of cysteine(Figure3.5). In place of a sulfur atom in the cysteine, Selenocysteine contains selenium (Se). Several human proteins and enzymes are seleno proteins. Selenocysteine is located in the active sites of enzymes that participate in oxidation–reduction reactions.

117) What is fibrinogen?

Fibrinogen is a glycoprotein complex, made in the liver, that circulates in the blood of all vertebrates. During tissue and vascular injury, it is converted enzymatically by thrombin into fibrin and then to a fibrin-based blood clot. Fibrin clots function primarily to occlude blood vessels to stop bleeding.

118) What is the function of sorbitol?

Sorbitol is a sugar alcohol found in fruits and plants with diuretic, laxative and cathartic property. Unabsorbed sorbitol retains water in the large intestine through osmotic pressure thereby stimulating peristalsis of the intestine and exerting its diuretic, laxative and cathartic effect. Sorbitol is also used as sweetener.

119) What is the sucrase location and function?

The sucrase-isomaltase enzyme is found on the surface of the intestinal epithelial cells, which are cells that line the walls of the intestine. These cells have fingerlike projections called microvilli that absorb nutrients from food as it passes through the intestine

120) Amino acids negatively charged (Acidic) R-group.

These amino acids are proton donor. At neutral PH. These amino acids are fully ionized, containing a negatively charged carboxylate group. Therefore, this amino acid are negatively charged at physiological PH.

121) Function of Amylase and Maltase

Amylase digest starch into smaller molecule yielding maltose, which in turn cleaved into two glucose molecules by maltase

122) Functions of Lipids in our body

In our body, lipid function as energy reserve, regulate hormone, transmit nerve impulses, and transmit fat soluble nutrients.

123) Beta conformation of proteins

The beta conformation organizes polypeptide chain in beta sheets. Beta sheets are second type of regular secondary structure that maximize hydrogen bonding between the peptide backbone.

124) Side chain properties of Serine and Threonine

Serine and threonine are aliphatic, polar, uncharged groups. The R group of these amino acid is more soluble in water or more hydrophilic than those of nonpolar amino acid because they contain functional groups that form hydrogen bond with water.

125) Alanine briefly explain and calculate its PI

Alanine contains both an alpha carboxyl and an alpha amino acid group at low pH both of these groups are protonated as the pH of solution raised the COOH group of form can dissociate by donating a proton to the medium the release of proton result in the formation of carboxylate group COO⁻ it is a dipolar molecule this is isoelectric point of alanine

126) Difference between L fucose and L rhamnose.

L-fucose is found in the complex oligosaccharide component to f-glycoprotein and glycolipid rhamnose is found in plant polysaccharide

127) Write rules of giving first letter name to amino acids.

If only one amino acid begins with particular letter, then that letter is used as its symbol example isoleucine = I If more than one amino acid begins with particular letter, the most common of these receives this letter as its symbols for example glycine is more common than glutamate so glycine = G. For the remaining amino acid, as one letter symbol is assigned that is as close in the alphabet as possible to the initial of amino acid

128) Explain amyloidosis

Amyloidosis is a generic term that refers to the extra cellular tissue deposition of low molecular weight of a variety of protein misfolded.

129) Muta-rotation

Muta-rotation is the change in optical rotation because of change in the equilibrium between two anomers.

130) Complex lipids and its three types

Ester of fatty acid containing groups in addition to an alcohol and fatty acid. These include phospholipids, glycolipid

131) What is native protein?

Protein which are in their folded and assembled form and are fully functional. they are unaltered by denaturing agent such as heat, chemical, enzyme action or other exogenesis of extraction.

132) Calculate PI of histidine?

$$PI = \frac{PK1 + PK2}{2}$$

$$PK1 = 2.3 \text{ (for carboxyl group)}$$

$$PK2 = 9.1 \text{ (for amino group)}$$

$$PI = \frac{2.3 + 9.1}{2} = 5.7$$

133) Define buffer?

Buffer is a substance that resist change in Ph when a small amount of acid is base is added to it.

134) Properties of lipids?

- Lipids may be either liquid or non- crystalline solid at room temperature.
- Pure fats and oils are colorless, odorless, and tasteless.
- They are insoluble in water, soluble in organic solvent.
- They are energy rich organic molecule

135) Formation of 2-deoxy-D-ribose from the D-ribose?

When oxygen of a OH group is removed in a monosaccharide then de-oxy sugars are formed 2 deoxy D-ribose is derived from D-Ribose and important constituent of DNA.

136) How light produced fireflies

The light produced by fireflies is the result of reaction involving the protein luciferin and Atp, catalyzed by enzyme luciferase.

137) How glycosidic bond is formed?

Glycosidic bond is formed by the condensation of alcohol or amine of one molecule and anomeric carbon of the sugar of other.

138) Contractile and transport proteins

Contractile protein is involved in muscle contraction and relaxation. Myosin of thick filaments. Transport protein transport various substances from one part of the body to other e.g. hemoglobin transport oxygen from lungs to tissue and carbon dioxide from tissue to lungs.

139) Name two proteins that contain selenocysteine? write their function of selenocysteine.

Serine and pyrrolysine are selenocysteine protein, selenocysteine is an L -alpha amino acid including certain peroxidase and reductase where it participates in the catalysis of electron transfer reaction.

140) Function of amide group in asparagine?

Asparagine is the amide of aspartic acid. It has high propensity to hydrogen bond, since the amide group can accept two and donate two hydrogen bond.

141) D and L isomerism

The designation of sugar isomer as the D-form or as L-form is determined by its spatial relationship to the parent compound of carbohydrate, the glyceraldehyde. The orientation of the -Hand-OH groups around the carbon atom adjacent to the terminal primary alcohol or carbon most distant from the carbonyl carbon the penultimate carbon. Carbon 5 in glucose determine whether the sugar belongs to D or L series when the -OH group in this carbon is on the right the sugar is D-isomer. When it is on the left it is L isomer

142) Octet rule problem

Octet rule cannot be used for type II cation .the physiological oxidation state of an iron atom has two positive charge instead of three charge ,and it's too large to fit into the plane of protoplasm however ,when iron is oxidized from ferrous ion to ferric ion there is loss of one extra electron ,forces between electron and proton decrease so that electron cloud penetrate more toward the nucleus as a result the ferric ion has smaller size then ferrous ion and fit into the proto-porphyrin plane.

143) Properties of non-polar aliphatic R groups.

- These are hydrophobic. these amino acids contain non polar side chain that do not bind, or give off proton.
- They are oily lipid like and promote hydrophobic interaction.
- These side chain cluster together in the interior of protein.
- This class include glycine, alanine, proline, valine leucine.

144) Messenger RNA Function

The messenger RNA functions as messengers to convey the information in a gene to the protein synthesizing machinery. The mRNA carries genetic information from the nuclear DNA to the cytosol, where it is used as a template for protein synthesis.

145) Physical properties of TAGs

- The simplest lipids constructed from fatty acids are the triacylglycerols, also referred to as triglycerides, fats, or neutral fats or storage lipids.
- Triacylglycerols are composed of three fatty acids in ester linkage with a single glycerol.

146) Order of reaction (enzyme)

When $[S]$ is much less, then the velocity of the reaction is approximately proportional to the substrate concentration. The rate of reaction is then said to be first order with respect to substrate. When $[S]$ is much greater than K_m the velocity is constant and equal to V_{max} . The rate of reaction is then independent of substrate concentration, and is said to be zero order with respect to substrate concentration.

Reaction Orders with Respect to Substrate Concentration

K_m equation Michaelis-Menten equation, the rate equation $V_o = V_{max} [S] / K_m + [S]$

- Where V_o = initial reaction velocity
- V_{max} = maximal velocity
- K_m = Michaelis constant = $K_1 + K_2 / K_1$
- $[S]$ = substrate concentration

This is the Michaelis-Menten equation-statement of the quantitative relationship between the initial velocity V_o , the maximum velocity V_{max} , and the initial substrate concentration $[S]$, all related through the Michaelis constant K_m .

An important numerical relationship emerges from the Michaelis-Menten equation in the special case when V_o is exactly one-half.

- $V_{max} V_o = V_{max} [S] / K_m + [S]$ $V_{max}/2 = V_{max} [S] / K_m + [S]$
- On dividing by V_{max} $1/2 = [S] / K_m + [S]$
- Solving for K_m , we get $K_m + [S] = 2[S]$ $K_m = [S]$, when $V_o = 1/2 V_{max}$

This is a very useful, practical definition of K_m K_m is equivalent to the substrate concentration at which V_o is one-half V_{max} .

147) Note on inorganic cofactors

Some enzymes require no chemical groups for activity other than their amino acid residues. Whereas some enzymes require molecules other than proteins for enzymatic activity. If the non-protein moiety is a metal ion such as Zn^{2+} or Fe^{2+} , it is called a cofactor. Some inorganic elements serve as co factor as enzymes such as Cu^{2+} cytochrome oxidase Fe^{2+} , Fe^{3+} cytochrome oxidase, catalase hydro peroxidase K^+ pyruvate kinase Mg^{2+} hexokinase, pyruvate 6 glucose phosphate, pyruvate kinase Mn^{2+} arginase, ribonucleotides reductase

148) Descriptive note on large and small regulatory RNA 10 marks.

large & Small Noncoding Regulatory RNAs One of the most exciting discoveries in the last decade of eukaryotic regulatory biology has been the identification and characterization of regulatory nonprotein coding RNAs (ncRNAs).

NcRNAs exist in two general size classes

- small consisting of microRNA (miRNAs) and silencing (siRNAs)
- large consisting of long noncoding RNAs (lncRNAs)

The small ncRNAs termed microRNA (miRNAs) and silencing (siRNAs) typically inhibit gene expression at the level of specific protein production by targeting mRNAs through one of several distinct mechanisms. Both siRNAs and miRNAs typically hybridize, via the formation of RNA–RNA hybridization to their targeted mRNAs

long noncoding RNAs (lncRNAs)

lncRNAs, which as their name implies, do not code for protein (i.e., the mRNA encoding genes). ncRNAs make up a significant portion of eukaryotic transcription ncRNAs play many roles ranging from contributing to structural aspects of chromatin to regulation of mRNA gene transcription by RNA polymerase II

149) How messenger RNA is heterogeneous class?

This class is the most heterogeneous in Abundance Size (500-6000 nucleotides) base sequence

- Stability
- mRNA comprise about 2–5% of total cellular
- RNA mRNA molecules are formed with the help of DNA template strand (3′-5′) during the process called transcription.

In addition to the protein coding regions in the mature eukaryotic mRNA that can be translated, there are untranslated regions at its 5′ and 3′ ends Moreover, there is a 5′ cap and a poly A tail at 3′ ends

150) Medical applications of nucleotide and nucleoside

Specifically, medical applications include the use of synthetic purine and pyrimidine analogs that contain halogens, thiols, or additional nitrogen atoms. Their use includes chemotherapy for cancer as suppressors of the immune response during organ transplantation. as anti-viral drugs such as in the treatment of AIDS

151) How activation energy is related to Order of reaction.

Activation energy ΔG

The difference between the energy levels of the ground state and the transition state is the activation energy ΔG .

The rate of a reaction reflects this activation energy higher activation energy corresponds to a slower reaction. Reaction rates can be increased by raising the temperature, thereby increasing the number of molecules with sufficient energy to overcome the energy barrier.

Alternatively, the activation energy can be lowered by adding a catalyst. Catalysts enhance reaction rates by lowering activation energies. The activation energy is lower when the enzyme catalyzes the reaction. The role of enzymes is to accelerate the inter-conversion of S and P. i.e., enzymes lower the energy of activation.

152) Function of second messenger nucleotides

Second messengers Nucleotides, such as cyclic AMP (cAMP) and cyclic GMP (cGMP), serve as second messengers in signal transduction pathways.

153) How iso enzymes of hexokinases illustrate the significance of km

A comparison between the isozymes of hexokinase illustrates the significance of the K_m . Hexokinase catalyzes the first step in glucose metabolism in most cells, the transfer of a phosphate from ATP to glucose to form glucose 6-phosphate. Hexokinase I, the isozyme in red blood cells has a low K_m for glucose of approximately 0.05 mM- helpful in utilizing blood glucose even when the blood glucose concentration is very low. The isozyme of hexokinase, called glucokinase, which is found in the liver has a much higher K_m of approximately 5 to 6 Mm - helpful in storing large amounts of "excess" glucose as glycogen or converting it to fat after a carbohydrate meal.

154) Bile salt

The emulsification of dietary fats in intestinal canal, brought about by bile salts, is a prerequisite for digestion and absorption of fats. The bile salts, act to break apart the fat globules in the small intestines and allow them to become more "soluble" for absorption. The hydrophobic fat molecules will clump together into globules in the watery mixture in the digestive system. The emulsifiers break them down to smaller "globules" and allow them to become more soluble.

155) Three basics steps of enzymes catalyzed actions

Enzyme-catalyzed reactions have three basic steps:

- **binding of substrate:** $E+S \leftrightarrow ES$
- **conversion of bound substrate to bound product:** $ES \leftrightarrow EP$
- **release of product:** $EP \leftrightarrow E+P$

To understand catalysis, we must first appreciate the important distinction between reaction equilibria and reaction rates.

156) Condensation reaction

N-glycosidic bond Sugars are linked to the heterocycle by a β -N-glycosidic bond, almost always to the N-1 of a pyrimidine N-9 of a purine. The N-glycosyl bond is formed by removal of the elements of water, a hydroxyl group from the pentose and hydrogen from the base. Thus it is a condensation reaction.

157) What is concentration of OH in a solution with an H Concentration What is the concentration of OH in a solution with an H⁺ concentration of $1.3 \times 10^{-4} \text{M}$?

Solution:

$K_w = [\text{H}^+][\text{OH}^-]$ With $[\text{H}^+] = 1.3 \times 10^{-4} \text{M}$,

solving for $[\text{OH}^-]$ gives $[\text{OH}^-]$

$= K_w / [\text{H}^+] = 1 \times 10^{-14} \text{M}^2 / 1.3 \times 10^{-4} \text{M}$

$= 7.7 \times 10^{-11} \text{M}$

158) Melting temperature T_m

When DNA is heated, the temperature at which one half of the helical structure is lost is defined as the melting temperature. A specific sequence of DNA has a characteristic denaturation temperature, or melting point (T_m)

159) Bee wax and spermaceti

Bees-wax is secreted by the honeybees that use it to form the combs. It is a mixture of waxes chief constituent is myricyl palmitate (30C) (16C) Spermaceti is a wax that is most often found in the head cavities of the sperm whale.

160) Saturation kinetics

The leveling off of the reaction rate at high substrate concentrations reflects the saturation with substrate of all available binding sites on the enzyme molecules present. This finite limit of V_{max} (maximum velocity) is called saturation kinetics. Saturation kinetics is a characteristic property of all rate processes dependent on the binding of a ligand to a protein.

161) Name two unnatural pyrimidine.

- Fluorouracil (5FU)
- 6-Aza Cytosine (AZC)

162) Name two unnatural Purine Bases

- Mercaptopurine
- Allopurinol
- 8-Azaguanine

163) Describe lipoxins

The lipoxins are formed through the action of 15-lipoxygenase followed by the action of 5-lipoxygenase on arachidonic acid. A series of reductions of the resultant hydro-per-oxy groups leads to the formation of trihydroxy derivatives of arachidonic acid known as the lipoxins. Lipoxins induce chemotaxis and stimulate superoxide radicals for killing of microorganisms

164) Name two second messenger of nucleotides?

1. Clandestine mono phosphate(cAMP)
2. Cyclic Gaunoadenosine mano phosphate(cGMP)

165) What are simple lipids? give two types of simple lipids. (2 marks)

A simple lipid is a fatty acid of different type of alcohol and carry no other substances. A simple lipid is belonging to the heterogeneous class of predominately non polar in nature. They are in soluble in water but soluble in organic solvent, such as chloroform and benzene.

Simple Lipids: Fatty acids ester of different alcohol. Fats: Fatty acid ester of different glycerol. Oil and fats are in liquid state. They are insoluble in water but soluble in organic solvent such as chloroform and benzene.

Waxes: Solid ester of fatty acid ester. They are insoluble in water due to weak polar nature of ester group.

166) Buffer Composition

To relatively the maintain the pH of the solution, a buffer is must consist the acid-base pair meaning either:

- i) A weak base and a conjugate acid.
- ii) A weak acid and conjugate base.

The use of one or more, depend upon the desired pH when preparing the buffer.

Example: Acetic acid such as sodium acetate (CH_3COOH) in which they have conjugate acid. Ammonia (NH_3) and a salt (NaCl) in which they have conjugate base.

167) Define saponification and saponification Number? (3marks)**Saponification number**

The number of milligrams of the sodium hydroxide and potassium hydroxide of free or combined state of fats or wax to completely saponifying are called Saponification number.

Saponification

“Hydrolysis of a fat by an alkali is called saponification”.

The resultant products are glycerol and the alkali salts of the fatty acids, which are called “soaps”. The number of mgs of NaOH/KOH required to saponify the free and combined FA in one gram of a given fat is called its saponification number The amount of alkali needed to saponify a given quantity of fat will depend upon the number of carboxylic ($-\text{COOH}$) group present. Thus, fats containing short chain fatty acids will have more $-\text{COOH}$ groups per gram than long- chain fatty acids and this will take up more alkali And hence will have higher saponification number

168) Define enzyme kinetics?

Enzymes are protein catalysts that, like all catalysts, speed up the rate of a chemical reaction without being used up in the process. They achieve their effect by temporarily binding to the substrate and, in doing so, lowering the activation energy needed to convert it to a product.

Enzyme kinetics is the study of the chemical reactions that are catalyzed by enzymes. In enzyme kinetics, the reaction rate is measured and the effects of varying the conditions of the reaction are investigated. Studying an enzyme's kinetics in this way can reveal the catalytic mechanism of this enzyme, its role in metabolism, how its activity is controlled, and how a drug or an agonist might inhibit the enzyme. Enzyme kinetics is the investigation of how substrate bind with enzyme then into product. They are used to kinetic analysis are commonly obtained from an enzyme.

In 1913 the Leonor Michalis and Maud Leonora Manton are proposed quantitative theory of enzyme. The molecules of the substrate bind reversibly with enzyme are called enzyme substrate complex. These molecules are converted into product are called enzyme product complex. These theories are further developed by J.B.S Halden and G.E Briggs who derived equation to still widely used today. Enzyme Kinetics depend upon the solution condition and saturation concentration.

169) Properties of waxes (5 marks)

- Waxes are insoluble in water
- soluble in fat solvents and are negative for acrolein test.
- very resistant to rancidity.
- Waxes are not easily hydrolyzed as the fats and are indigestible by lipases (enzymes responsible for fat digestion in body)
- they are of no nutritional value

170) Nucleotides as 'energy currency' of the cell? (5 marks)

Energy currency: Nucleotides play an important role as "energy currency" in the cell. Nucleoside tri- and diphosphates such as ATP and ADP are the principal donors and acceptors of phosphoryl group in metabolism. By doing this, they play a key role in the energy transduction. This energy is used in almost every energy requiring process in the body, such as; Muscle contraction, Transmission of nerve impulse, Transports of nutrients across cell membrane. Motility of spermatozoa and many more energy dependent processes.

171) Define enzymes? Write classification of enzymes? (10 marks)

Enzymes are protein molecules in cells which work as catalysts. Enzymes speed up chemical reactions in the body, but do not get used up in the process. Almost all biochemical reactions in living things need enzymes. With an enzyme, chemical reactions go much faster than they would without the enzyme. There were six classes of enzymes that were created so that enzymes could easily be named.

These classes are: 1) Oxidoreductases, 2) Transferases, 3) Hydrolases, 4) Lyases, 5) Isomerases, 6) Ligases.

This is the international classification used for enzymes.

1) Oxidoreductases

These catalyze oxidation reduction reactions and are further divided into four subgroups;

- Oxidase, • Dehydrogenases, • Hydro peroxidases • Oxygenase.

2) Transferases

These bring about a transfer of functional groups such as phosphate and amino group from one molecule to another molecule called donor and acceptor molecules respectively. The common examples of this group are Transminases, Phosphotransferases (Kinases), Hexokinase is a phosphotransferase which catalyze the transfer of phosphate groups .



3) Hydrolases

These enzymes catalyze hydrolysis, i.e., add water molecule to the substrate which is simultaneously decomposed; the functional group of substrates is transferred to water. Common example of hydrolases is Protein hydrolyzing Enzymes (peptidases), Carbohydase, Lipid hydrolyzing enzymes e.g., Lipases and Phospholipases.

4) Lyases

Theses enzymes catalyze the addition of NH_3 , H_2O or CO_2 to double bonds or the removal of these groups leaving behind double bonds. Lyases are included in a separate class because they catalyze these reactions by means other than hydrolysis or oxidation.

5) Isomerases

These enzymes catalyze the structural change within a single molecule by the transfer of groups within it, resulting in the formation of an isomeric form of substrate.

6) Ligases

These enzymes catalyze condensation reactions joining two molecules by forming

- C-O
- C-S
- C-N
- C-C bonds.

The energy for condensation is provided by cleavage of high energy phosphates, e.g., ATP, GTP etc.

172) Components of nucleotide

Nucleotides are composed of

- A nitrogenous base (purine or pyrimidine)
- A pentose monosaccharide
- One, two, or three phosphate groups

173) Characteristics of VLDL .

VLDLs are assembled in the liver. Composed predominantly of TAGs synthesized in liver and contain some cholesterol and cholesteryl esters. As VLDL pass through the circulation, TAG is degraded and taken up by peripheral tissues in the form of fatty acids, causing the VLDL to decrease in size and become denser, called VLDL remnant.

174) What is lipoproteins

A lipoprotein is a biochemical assembly whose primary purpose is to transport hydrophobic lipid (a.k.a. fat) molecules in water, as in blood or extracellular fluid. They have a single-layer phospholipid and cholesterol outer shell, with the hydrophilic portions oriented outward toward the surrounding water and lipophilic portions of each molecule-oriented inwards toward the lipid's molecules within the particles.

175) Five functions of cyclic AMP

- Acts as second messenger in the cell. It has role in glycogen metabolism cAMP, glycogenolysis, cAMP TAG metabolism, cAMP lipolysis
- It decreases cholesterol synthesis
- It causes activation of protein kinases which in turn activate or deactivate other enzymes.
- It regulates the cell membrane permeability, by increasing permeability of cell membrane to H₂O, Na⁺, K⁺ & Ca²⁺
- Moreover, it regulates insulin secretion, catecholamine biosynthesis & Melatonin synthesis
- Cyclic GMP is synthesized from GTP
- It serves as a second messenger in response to nitric oxide during relaxation of smooth muscle (especially blood vessels) so it has role in smooth muscle relaxation and vasodilatation.
 - It also has role in Protein Phosphorylation Neurotransmission, Insulin action, Regulation of sodium channels

176) Define glycerol

Glycerol is widely used in pharmaceutical and cosmetic preparations.

It has the following properties:

- Colorless
- Viscous oily liquid with sweet taste.

177) Two properties of glycerol trinitrate.

Glycerol combines with three molecules of nitric acid to form Glycerol trinitrate that is used as

- explosive
- vasodilator

178) Name of Three pyrimidine bases.

Pyrimidines include:

- Cytosine (C)—in both DNA and RNA
- Thymine (T)—only in DNA
- Uracil (U) —only in RNA

179) Write five Properties of nitrogenous bases.

- **Aromatic:** The Nitrogen containing bases are aromatic i.e. they have alternate double bonds
- **Heterocyclic:** They are heterocyclic i.e. structures that contain other atoms in addition to carbon, such as nitrogen in the ring structure. The six-atom rings of purines and pyrimidines are numbered in opposite directions.
- **Weak Bases:** Purines or pyrimidines with an -NH_2 group are weak bases
- **Functional Groups:** The most important functional groups of pyrimidines and purines are • ring nitrogen's • carbonyl groups • exocyclic amino groups
- **Hydrophobicity:** The purine and pyrimidine bases are hydrophobic and relatively insoluble in water at the near-neutral cell pH
- **Stacking Interaction:** Hydrophobic stacking interactions in which two or more bases are positioned with the planes of their rings parallel (like a stack of coins) are one of two important modes of interaction between bases in nucleic acids. Base stacking helps to minimize contact of the bases with water, and these interactions are very important in stabilizing the three-dimensional structure of nucleic acids

180) Composition of triacylglycerol

The simplest lipids constructed from fatty acids are the triacylglycerols, Also referred to as; triglycerides, fats, or neutral fats or storage lipids.

Triacylglycerols are composed of three fatty acids in ester linkage with a single glycerol

181) Carbon atom number in pentose sugar of nucleotides & nucleosides

The carbon and nitrogen atoms in the rings of the base and the sugar are numbered separately. The atoms in the rings of the bases are numbered 1 to 6 in pyrimidines & 1 to 9 in purines.

In the pentoses of nucleotides and nucleosides the carbon numbers are given a prime (') designation to distinguish them from the numbered atoms of the nitrogenous base.

The carbons in the pentose are numbered 1' to 5'. Numerals with a prime (e.g., 2' or 3') distinguish atoms of the sugar from those of the heterocycle. Thus, when the 5'-carbon of a nucleoside (or nucleotide) is referred to, a carbon atom in the pentose, rather than an atom in the base, is being specified.

182) Nucleotide serve as single transduction pathway?

Signal Transduction: GTP and GDP play key roles in activating or inhibiting proteins in various cellular signaling cascades.

Medical applications

Specifically medical applications include the use of synthetic purine and pyrimidine analogs that contain halogens, thiols, or additional nitrogen atoms.

There use include chemotherapy for cancer as suppressors of the immune response during organ transplantation. as antiviral drugs such as in the treatment of AIDS

183) Rancidity and its factors

It is a physico-chemical change in the natural properties of the fat leading to the development of unpleasant odor or taste or abnormal color. It occurs particularly on aging after exposure to atmospheric oxygen, light, moisture, bacterial or fungal contamination and/or heat. Saturated fats resist rancidity more than unsaturated fats that have unsaturated double bonds.

- Rancidity is due to • Oxidation • Hydrolysis

Rancidity is a very general term and in its most general meaning, it refers to the spoilage of a food in such a way that it becomes undesirable (and usually unsafe) for consumption. When people say that a food has "gone bad," what they're usually talking about is rancidity.

Rancidity is the complete or incomplete oxidation or hydrolysis of fats and oils when exposed to air, light, moisture or by bacterial action, resulting in unpleasant taste and odor. ... When these processes occur in food, undesirable odors and flavors can result.

184) Name the pentose sugar in nucleic acids

D-ribose and 2-deoxy D-ribose are the only sugars so far found in the nucleic acids. These Aldo pentoses belong to D-family. They are present as Furanose (ring) in the form of β -Anomer

The addition of a pentose sugar to a base produces a nucleoside. If the sugar is D-ribose, a ribonucleoside is produced. If the sugar is 2-deoxy D- ribose, a deoxyribonucleoside is produced.

185) Halogenation with respect to fat

Similar to hydrogenation Halogens such as chlorine, bromine and iodine can also be added to double bonds in unsaturated fatty acids. It is a very important property to determine the degree of unsaturation of the fat or oil that determines its biological value. The degree of unsaturation is reflected by Iodine number.

- Iodine number is defined as the number of grams of iodine absorbed by 100 gm of fat.
- The more the iodine number, the greater the degree of unsaturation.

Fats rich in saturated fatty acids have low iodine numbers, while fats rich in unsaturated fatty acids have high iodine numbers. The determination of iodine number is useful to the chemist in determining the quality of an oil or its freedom from adulteration. Iodine number of cotton seed oil varies from 103 to 111. That of olive oil from 79 to 88, And that of linseed oil from 175 to 202. A commercial lot of olive oil which has iodine number higher than 88 might have been adulterated with cotton seed oil. The higher is the iodine number, the more reactive, less stable, more susceptible to oxidation and rancidification is the oil or fat.

186) Nucleoside vs. Nucleotide

- A nucleoside consists of a nitrogenous base covalently attached to a sugar (ribose or deoxyribose) but without the phosphate group.
- A nucleotide consists of a nitrogenous base, a sugar (ribose or deoxyribose) and one to three phosphate groups.

Nucleoside = Sugar + Base

Nucleotide = Sugar + Base + Phosphate

Nucleotide is composed of a nitrogenous base, sugar and a phosphate group.

Nucleoside: Nucleoside is composed of only a nitrogenous base and a phosphate group

187) iodine number

Iodine number is defined as the number of grams of iodine absorbed by 100 gm of fat. The more the iodine number, the greater the degree of unsaturation. Fats rich in saturated fatty acids have low iodine numbers, while fats rich in unsaturated fatty acids have high iodine numbers

188) example of condensation reaction

Sugars are linked to the heterocycle by a β -N-glycosidic bond, almost always to the N-1 of a pyrimidine to N-9 of a purine. The N-glycosyl bond is formed by removal of the elements of water (a hydroxyl group from the pentose and hydrogen from the base)

Thus, it is a condensation reaction. Similar to O-glycosidic bond formation in carbohydrates

However, N-glycosidic bonds, have Nitrogen atom instead of oxygen linking the two residues. the addition of the glycosidic bond to nitrogenous base is indicated by the name change such as from adenine to adenosine for the glycosidic bond

189) Steroids and Cholesterol

A steroid is a lipid whose structure is based on the tetracyclic (four-ring) structure consists of

- 3 cyclohexane rings.
- 1 cyclopentane ring.

Steroids with eight to ten carbon atoms in the side chain at C-17 and a hydroxyl group at C-3 are classified as sterols

190) Acrolein Test

Acrolein test is used to detect the presence of glycerol or fat. When fat is treated strongly in the presence of a dehydrating agent like potassium bisulphate (KHSO₄), the glycerol portion of the molecule is dehydrated to form an unsaturated aldehyde, Acrolein that has a pungent irritating odor.

191) Hydrogenation of Fats

Hydrogenation converts liquid vegetable oils into solid or semi-solid fats, such as those present in margarine. Changing the degree of saturation of the fat changes some important physical properties, such as the melting range, which is why liquid oils become semi-solid

192) Difference Between Thymine and Uracil?

Thymine becomes thymidine and deoxy thymidine,³¹ Thymine is 2,4-dioxy-5-methyl -pyrimidine, Thymine(T)—only in DNA.

Uracil(U) becomes uridine and deoxy uridine, Uracil is 2, 4- deoxy pyrimidine, Uracil (U)—only in RNA.

T and U differ by only one methyl group, which is present on T but absent on U.

193) Palmitic acid main carbons.

Ans: 16 carbons are present in palmitic acid

194) How banaspati gee is produced from oil.

Bans Patti Ghee is made with the combination of Hydrogenated Oils, Refined Sesame Oil. The ghee is manufactured state of the art multi stage hygienically controlled plant. It gives a mouth-watering aroma and taste to your food. It is low in cholesterol which makes it safer for the heart patients. Vitamin A and D are added.

195) Name factors that contract entropy.

Here are some situations in which entropy increases:

- The entropy increases whenever heat flows from a hot object to a cold object.
- It increases when ice melts, water is heated, water boils, and water evaporates.
- The entropy increases when a gas flows from a container under high pressure into a lower pressure.
- It increases when you spray something out of an aerosol can or you let air out of a tire.

196) Galactosyl ceramidase

Galactosyl ceramide is a major glycol sphingo lipid of brain and other nervous tissue

- It contains a number of characteristic C24 fatty acids
- eg, cerebronic acid

Globosides are glycol sphingolipids with two or more sugars, usually D -glucose, D- galactose, or N-acetyl-D-galactosamine.

197) Define ammolite.

A semi-precious gem stone consisting of fossilized ammonite shells. The winner will be sent an ammolite trophy, the official gem stone of the Alberta

198) Carrier of intermediate.

Intermediate carrier definition: is a transportation line participating in a through movement which neither originates nor terminates the passengers or freight

199) Importance of sorbitol.

Sorbitol can be used as a laxative when taken orally or as an enema. Sorbitol works as a laxative by drawing water into the large intestine, stimulating bowel movements. Sorbitol is a low-calorie sweetener chemically extracted from glucose. It is used as an alternative to sugar in a range of foods, including low calorie.

200) Properties of glycerol?

- Glycerol is a colorless, hygroscopic, highly viscous syrupy liquid.
- It is miscible with water and alcohol in all proportions.
- The boiling point of glycerol is 563 K which is quite high.
- Glycerol is non-toxic in nature.

201) Diseases caused by the deficiency of enzymes?

Various enzyme deficiencies inside lysosomes can result in build-up of toxic substances, causing metabolic disorders including:

- Hurler syndrome (abnormal bone structure and developmental delay)

- Niemann-Pick disease (babies develop liver enlargement, difficulty feeding, and nerve damage)

202) **Structure of amino acids?**

An amino acid is an organic molecule that is made up of a basic amino group ($-NH_2$), an acidic carboxyl group ($-COOH$), and an organic R group (or side chain) that is unique to each amino acid. The term amino acid is short for α -amino [alpha-amino] carboxylic acid.

203) **What is Complex lipid and describe its types?**

Complex Lipids

Complex lipids are esters of fatty acid containing group in addition to an alcohol and a fatty acids.

Types of Complex Lipids

Generally complex lipids are of three types namely phospholipids, glycolipids and lipoamino acids.

- **Phospholipids:**

Phospholipids (PL) are a group of polar lipids that consist of two fatty acids, a glycerol unit and a phosphate group which is esterified to an organic molecule such as choline, ethanolamine, inositol, etc.

- **Glycolipids:**

Glycolipids are lipids with a carbohydrate attached by a glycosidic (covalent) bond such as glycosylated sterols, gangliosides etc.

OR

A lipid such as a ganglioside or a cerebroside that contains a carbohydrate radical.

- **Lipoamino acids:**

Any class of lipid containing amino acid residues with or without glycerol or fatty acid residues but lacking a phosphate group that are present in bacteria and similar organisms.

204) **Name of 22nd amino acid?**

Pyrrolysine (Pyl, O) is a genetically encoded amino acid that occurs naturally in methanogenic archaea, now designated the 22nd amino acid.

205) **Formation of lysine?**

Lysine and other amino acids are commonly produced by fermentation using strains of heterotrophic bacteria, such as *Escherichia coli* and *Corynebacterium glutamicum*.

206) **How gluconic acid of glucose is formed?**

Gluconic acid (pentahydroxycaproic acid) is produced from glucose through a simple dehydrogenation reaction catalyzed by glucose oxidase.

207) What is Rhamnose?

Rhamnose is a naturally occurring deoxy sugar. It can be classified as either a methyl-pentose or a 6-deoxy-hexose. Rhamnose predominantly occurs in nature in its L-form as L-rhamnose. This is unusual, since most of the naturally occurring sugars are in D-form.

208) What is the cooperative binding of hemoglobin? (3)

The protein cannot be considered in terms of four independently oxygen-binding subunits. As hemoglobin binds successive oxygen, the oxygen affinity of the subunits increases. The affinity for the fourth oxygen to bind is approximately 300 times that for the first.

209) Write two examples of Cyclic Nucleotide?

- Cyclic AMP (cAMP)
- Cyclic GMP (cGMP)

210) Write three functions of CAMP?

cAMP is a second messenger, used for intracellular signal transduction, such as transferring into cells the effects of hormones like glucagon and adrenaline, which cannot pass through the plasma membrane. It is also involved in the activation of protein kinases.

211) Difference between oxidative and hydrolytic rancidity?**Hydrolytic rancidity**

Hydrolytic rancidity refers to the odor that develops when triglycerides undergo hydrolysis and the release of the free fatty acids.

Oxidative rancidity

Oxidative rancidity is the chemical reaction of the oil with oxygen.

212) What do you know about K_m of Micheals Menten Equation?

K_m is the Michaelis-Menten constant which shows the concentration of the substrate when the reaction velocity is equal to one half of the maximal velocity for the reaction.

213) Role of Nucleotides as coenzyme and intermediate carrier?**As Coenzyme:**

Several nucleotides are coenzymes; they act with enzymes to speed up (catalyze) biochemical reactions. The nitrogen-containing bases of nearly all nucleotides are derivatives of three heterocyclic compounds: pyrimidine, purine, and pyridine.

As Intermediate carrier:

In nucleotides like ATP, the energy is stored in the phosphor-anhydride bonds between the three phosphate groups. These bonds require energy to form, so the cells in your body can use these bonds like a battery to store extra power.

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